Methods 02: Using Claude Code as a tutor with the notebook

MPhil Past and Future Climates · Computational techniques for paleoclimate research · Francesco Muschitiello, University of Cambridge

Claude is **optional**. Use it to understand the notebook and to try small changes, not to do the thinking for you. You learn most when you predict first and then check.

> **Before you start,** finish the setup guide *Methods 02: Set up Python and VS Code*. The notebook must already run with the `pfc` kernel.

Install the extension and sign in

This is Option 2 in the Methods 1 guide *Getting_Started_with_Claude.pdf*. In short:

1. You need **VS Code 1.94 or later** (**Code > About Visual Studio Code** on a Mac; **Help > About** on Windows).
2. Open the Extensions view (`Cmd+Shift+X` (Windows: `Ctrl+Shift+X`)), search for **Claude Code** (publisher: Anthropic) and click **Install**.
3. Open a file, then click the **Spark icon** at the top right of the editor, or **Claude Code** in the status bar at the bottom.
4. Click **Sign in** and approve in your browser with the claude.ai account from the course Team invitation.

> **Three things that stop it working.** (1) The extension does not work in **Restricted Mode**: trust the folder. (2) To switch to a different account, type `/logout` in the Claude panel and sign in again. (3) If `ANTHROPIC_API_KEY` is set on your computer, it can override your login. Unset it, restart VS Code, and ask the instructor if you are not sure how.

Set up before you ask (2 minutes)

1. **Pick the kernel first.** Open the notebook and choose `pfc` at the top right.
2. **Work on a copy.** Right-click the notebook > **Copy**, **Paste**, then rename it (for example `Methods_02_yourname.ipynb`). Open the copy.
3. **Save** before you ask: `Cmd+S` (Windows: `Ctrl+S`). Claude then sees your latest version.
4. **Open the Claude panel** (Spark icon).
5. **Set the mode to Manual** (or Plan). Look for the mode indicator below the prompt box and click it.

> **Newer versions start in 'Auto' mode.** Auto edits files and runs commands without asking you each time. Switch to **Manual**, which asks before every change, or **Plan**, which only writes a plan.

How to ask

- **Select** the code or text you are asking about. The footer of the Claude panel shows how many lines Claude can see.
- Press `Option+K` (Windows: `Alt+K`) to insert an **@-reference** to the selection into your prompt.
- Selecting text inside notebook cells is **not guaranteed** to reach Claude the same way as in an ordinary script. If Claude answers about the wrong thing, type `@` and the notebook name (for example `@Methods_02_yourname.ipynb`), or paste the code into the prompt.

Good prompts (sections of this notebook in brackets)	Not good
“Explain this cell in plain English. I have never used Python.”	“Do Task 3 for me.”
“Why is the correlation for this coral negative?” (Part 2)	“Make it better.”
“What does this spectrum figure show?” (section 2g)	“Install this package.”
“I get this error: *(paste it)*. What does it mean?”	“Write my conclusions.”
“In this cell, make the dots on the map bigger.” (section 1e)	“Fix everything.”
“Add a line that prints how many corals are left.” (section 1d)	

Adapt the code: one small change at a time

1. **Ask for one change in plain words.** For example: “In this cell, make the dots on the map bigger.”
2. **Read the diff.** Claude shows the change before it is made: red lines are removed, green lines are added. Click **Accept** or **Reject**. You can also edit the text before you accept. If you do not understand the change, reject it and ask “Explain this change”.
3. **Accept, save** (`Cmd+S` (Windows: `Ctrl+S`)) **and run the cell**. Check that the figure or number changed the way you expected.
4. **Something broke?** Hover over your message in the Claude panel and choose **Rewind code to here**. Or delete your copy and make a new one from the original.
5. After several edits, use **Restart** and then **Run All** (toolbar at the top of the notebook) to check that it still runs from the top.

> **Rewind has limits.** It does not undo changes made by terminal commands (for example `pip install`), and it is not a replacement for version control. Your copy of the notebook is your safety net.

Bigger ideas: use Plan mode

For something larger, such as “compare 1900–1940 with 1940–1980”, switch the mode to **Plan** first. Claude writes a plan and waits. Read it, ask questions, and only then let it make changes, one step at a time.

Six rules

1. **Predict first.** Say what you expect before you ask. Afterwards, say what the figure shows in your own words.
2. **Copy only.** Never let Claude edit the original notebook.
3. **No installs.** If Claude proposes `pip install` or `conda install`, click **Reject** and ask the instructor. If Claude asks to **Execute** a cell in your notebook, read the cell first and click **Cancel** if it installs anything (for example `%pip install`).
4. **Check the science.** Claude can be wrong. Compare its answers with the notebook text and the papers cited in it.
5. **Usage limits are personal** to your account. When you reach yours you see “You've hit your session limit”. Type `/usage` in the panel to see how much you have used. Carry on with the notebook on your own.
6. **No personal data.** Do not paste personal or private information into the prompt.

Troubleshooting

Problem	What to do
No Spark icon	Open a file in a trusted folder first. If it is still missing, press `Cmd+Shift+P` (Windows: `Ctrl+Shift+P`) and run **Developer: Reload Window**.
Claude says a package is missing	Pick the `pfc` kernel in the notebook first, then try again. Claude's own terminal may not use your environment: do **not** let it install anything.
Signed in with the wrong account	Type `/logout` in the Claude panel, then sign in again with your course account.
The notebook looks odd after an edit	Save, close the notebook and open it again.
Claude talks about the wrong cell or file	Type `@` and the notebook name, or paste the code into the prompt.
Anything else	Copy the whole message and post it in the course channel, saying which operating system you use.

Instructions checked on 19 October 2026 and may change as VS Code and its extensions update. Menu and button names in the Claude Code extension change between versions; if a name differs, look for the closest match.